
ImageAuditor: Membership Inference Attack against Image-based Retrieval-Augmented Generation

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Abstract

Image-based Retrieval-Augmented Generation (IRAG) conditions a frozen generator on reference images retrieved from an external database, supporting both text-to-image (T2I) and question answering (Q&A) tasks. Because these databases are opaque and typically web-scraped, copyright holders need ways to audit whether specific images appear in them. While prior work employs membership inference attacks (MIAs) to audit uni-modal, text-based RAG, they fail to transfer to IRAG due to *two* fundamental challenges. First, *cross-modal retrieval*: text-RAG MIAs force retrieval of the target passage by injecting its content into the query, which is unavailable in IRAG since images cannot be embedded into text queries; even accurate image captions fail to bridge the modality gap. Second, *discriminative signal extraction*: text-RAG MIAs extract membership signals by prompting the generator to answer multiple questions over the target passage, whereas T2I generators in IRAG produce images rather than follow Q&A commands. To fill this gap, we introduce the first MIA tailored to IRAG, *ImageAuditor*, which decomposes each attack query into a retrieval segment and an extraction segment, enabling dedicated optimization for each challenge. For retrieval, we propose Reward-Guided Policy Optimization (RGPO), which updates a stochastic policy from reward-ranked candidates to navigate the cross-modal embedding landscape and admits finite-sample optimality guarantees to balance exploration and exploitation. For extraction, we analyze the distribution of the MIA score to guide the co-design of the prompting strategy and scoring rule, and derive task-specific instantiations for T2I and Q&A tasks. We aggregate signals across queries via K-means clustering for reliable membership decisions. Across IRAG systems built on SDXL, SD1.5, Kandinsky, LLaVA-1.6, and Qwen2.5-VL, ImageAuditor exceeds 80% AUROC with only four queries per audited image and remains robust across diverse settings.

1 Introduction

Image-based Retrieval-Augmented Generation (IRAG) [22, 6] grounds generative models in external visual knowledge by retrieving reference images from an image database. It has gained increasing popularity in two major multimodal generation tasks (Figure 1): text-to-image generation (T2I) [22, 31, 5], where retrieved images supply concepts, identities, and styles beyond the reach of base diffusion models [26]; and question answering (Q&A) [6, 4, 18], where retrieved images provide knowledge-dependent visual evidence for vision-language models [37]. For consistency, this paper focuses on IRAG systems that take text as input and produce either images (T2I) or text (Q&A).

This rapid adoption also exposes a privacy and copyright surface [21, 3]. Because these databases are typically web-crawled with poorly documented provenance, copyright holders (e.g., artists), dataset owners, and privacy regulators have no reliable way to verify whether their images have been silently incorporated into a deployed system, motivating the need for principled auditing tools.

Membership inference attacks (MIA) [32], which infer whether a record belongs to a target system, provide a natural auditing primitive. Traditional MIAs [32, 1] target a model’s training data by exploiting memorization signals in its weights, and therefore do not apply when membership resides in an external database. Recent text-RAG MIAs [23, 16, 20] probe the database via question–answer interactions, but they implicitly assume *single-modal retrieval*, where the attacker can inject the target document into the query to force retrieval, and *text-format outputs*, where responses can be matched against pre-computed answers. Both assumptions break down in IRAG settings. A text query cannot carry an image target, and we show in Section 4.2 that even substituting a strong caption fails to bridge the modality gap. Furthermore, IRAG for T2I generation emits images rather than text and does not explicitly answer the query, leaving Q&A-style probing without a signal to match. We formalize these obstacles as *two fundamental challenges* in Section 3.1.

Contributions To fill this gap, we introduce *ImageAuditor*, the first MIA tailored to IRAG, operating in a realistic setting with only a handful of attack queries per audited image (Section 2.2). Our key design decomposes each query into two functionally isolated segments: a *retrieval segment* that brings the test image into the generator’s context when present, and an *extraction segment* that elicits a discriminative MIA signal from the resulting output. This decomposition allows each segment to be explicitly optimized against a clean and dedicated objective, avoiding conflicts inherent in coupled optimization.

For the retrieval segment, a natural starting point is the suffix-optimization toolkit developed for RAG *poisoning* attacks [43, 39], which consider a structurally similar problem of crafting text that retrieves a chosen target. However, their gradient-based optimizers are designed for single-modal text retrieval and are easily trapped in local minima on the non-smooth cross-modal embedding landscape, motivating a gradient-free, exploration-heavy alternative. We therefore propose *Reward-Guided Policy Optimization* (RGPO), which updates a stochastic token policy from reward-ranked candidates to balance exploration and exploitation, and comes with a finite-sample optimality guarantee (Theorem 3.1 in Section 3.2).

For the extraction segment, we derive a mixture-identity decomposition of the MIA score that cleanly separates retrieval probability from the extraction hit/miss gap, and use it to guide task-specific instantiations for T2I and Q&A (Section 3.3). We further pair these instantiations with K-means aggregation across queries (Section 3.4). Across IRAG systems built on six generators and diverse datasets, ImageAuditor consistently outperforms adapted baselines, exceeding 80% AUROC and reaching around 50% TPR@5%FPR with only four queries per audited image. These values are well above the bar for a strong MIA [1], establishing ImageAuditor as a practical auditing tool.

2 Background and Threat Model

2.1 Background

IRAG. We describe an IRAG system as a tuple $S = (R, D, G)$ consisting of a retriever R , an image database D , and a generator G , where each image $I \in D$ is encoded by an embedding model E that R uses for retrieval. The key idea is to leverage external image knowledge (e.g., web-crawled images) to compensate for knowledge gaps in the generator’s parameters. In this work, we cover IRAG for both T2I and Q&A tasks. Formally, given a text query T , the system retrieves the top- K relevant images and produces output y (an image for T2I, a text answer for Q&A).

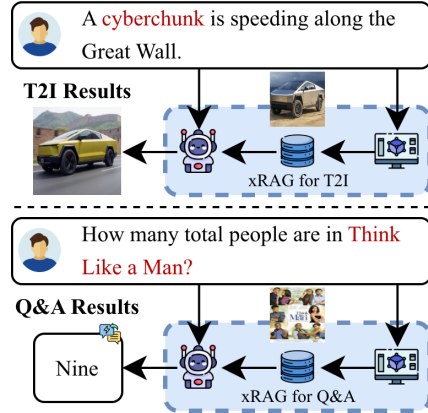


Figure 1: IRAG employs an external image database to generate faithful images (top) and produce reliable answers to knowledge-dependent queries (bottom).

Membership inference attack (MIA). MIA [32], which aims to determine whether a record belongs to a target database, is a prominent framework for auditing data privacy in ML systems. Prior MIAs [1, 12, 24] target a model’s training data and do not transfer to IRAG systems, where the target data must first be retrieved before it can influence the output. While recent work [20, 23] explores MIAs against text-based RAG, their query designs fail to reliably retrieve the target image due to the modality gap, especially under the small retrieval budgets used in IRAG (see Sec. 3.1). Moreover, their Q&A-style attack strategies do not transfer to IRAG systems whose generator is a text-to-image model. A separate line of work [3, 21] addresses IRAG copyright concerns from the *defender’s* side, embedding watermark patterns into images so that owners can later assert ownership. These approaches require proactive modification of the protected images and do not help general auditors (e.g., ordinary users) whose published data lacks such sentinels.

2.2 Threat Model

Attacker’s goal. The adversary acts as a data auditor (e.g., a copyright holder, a database owner, or a privacy regulator). Given a test image I and a target IRAG system S_{tgt} , the attacker aims to infer the membership $m \in \{0, 1\}$ of I in the image database D_{tgt} . We frame this as a binary classification task (member vs. non-member) intended to detect unauthorized use of copyrighted images.

Attacker’s capabilities. We consider a realistic and challenging attack setting. The attacker can only interact with the target IRAG system S_{tgt} through its API, operate under a limited query budget (e.g., 4–5 per test image) to remain stealthy, and has no knowledge of the images in D_{tgt} . For transparency, IRAG systems typically disclose the names of their AI components, such as the embedding model and the generator. Following the standard threat model in prior RAG attacks [39, 43, 10, 35, 2], we assume by default that the attacker has access to E_{tgt} . This assumption is realistic because (i) deployed IRAG systems overwhelmingly adopt off-the-shelf CLIP variants [27] as their retrievers [21, 31, 3, 6, 22, 5], since training a custom cross-modal encoder is expensive and rarely improves retrieval quality; and (ii) commercial providers further expose embedding APIs that return query embeddings directly.¹ Concretely, the attacker submits an image or text query and obtains its embedding via local deployment or the API. To stress-test this assumption, we additionally evaluate a restrictive scenario in which the attacker has no access to E_{tgt} and instead relies on a weaker shadow embedding model E_{sh} (e.g., a smaller model trained with a different architecture or on a different data distribution); ImageAuditor remains effective in this setting (Sec. 4.3). Finally, the attacker has access to a captioning model and, optionally, a small shadow dataset drawn from the same domain as the test images, which is readily collected from the Internet.

3 Method

3.1 Key Challenges

Designing an effective MIA against IRAG faces two fundamental challenges. **Challenge 1 (C1): Cross-modal retrieval.** Unlike traditional MIAs [32, 1], where members are memorized in model parameters and can be directly probed, our setting requires the test image to be retrieved into the generation context before it can influence the output of S_{tgt} . Crafting a text query that reliably retrieves a target image is non-trivial: text tokens are discrete, the embedding space is cross-modal, and the retrieval budget is small (e.g., 1–3). Prior text-RAG MIAs [23, 16, 20] sidestep this by injecting the target passage directly into the query—an option unavailable here, since an image cannot be embedded into a text query, and even a strong caption fails to bridge the modality gap (Sec. 4.2). White-box suffix-optimization attacks [39, 43] also suffer, as they are easily trapped in local minima on the non-smooth cross-modal landscape. **Challenge 2 (C2): Discriminative signal extraction.** Even when retrieval succeeds, the attacker must still coerce the black-box generator into producing outputs whose distributions diverge between members and non-members. This is non-trivial across modalities: T2I generators produce unstructured images rather than formatted answers, while Q&A generators tend to summarize multiple references and dilute the per-image signal. Prior Q&A-style strategies [23, 20] match generated text against pre-computed answers and require many queries per sample, which is incompatible with T2I outputs and impractical under realistic stealth budgets.

¹For example, Google’s multimodal embedding model `gemini-embedding-2` is accessible via API.

To address these issues, the optimization should both *explore* the embedding space and *exploit* high-reward regions within it. Motivated by recent approaches that leverage RL for prompt optimization [8, 41], we propose **Reward-Guided Policy Optimization (RGPO)**, which models the retrieval segment as a factorized stochastic policy $\pi(T_{\text{ret}} \mid T_{\text{ext}}) = \prod_{l=1}^L \pi_l(T_{\text{ret}}[l] \mid T_{\text{ext}})$, where π_l is the token distribution at position l . At iteration t , RGPO samples B candidates $\{T_{\text{ret}}^{(i)}\}_{i=1}^B \sim \pi_t$, scores them by the contrastive reward $r_i = f(T_{\text{ret}}^{(i)}; I, T_{\text{ext}})$, and assigns each a sharpened, elite-gated weight $w_{t,i} = \exp(r_i/\tau) \mathbf{1}\{r_i \geq \gamma_t\}$, where γ_t is the $(1 - \rho)$ -quantile of $\{r_i\}$ and $\tau > 0$ is a temperature. For each position l , the marginal is then updated toward the reweighted empirical token frequencies:

$$\pi_{t+1,l}(v \mid T_{\text{ext}}) = (1 - \beta) \pi_{t,l}(v \mid T_{\text{ext}}) + \beta \sum_{i=1}^B \tilde{w}_{t,i} \mathbf{1}\{T_{\text{ret}}^{(i)}[l] = v\},$$

with $v \in \mathcal{V}$ a vocabulary token, $\tilde{w}_{t,i} = w_{t,i} / \sum_j w_{t,j}$, and $\beta \in (0, 1]$ controlling the adaptation speed. Intuitively, RGPO shifts probability mass toward tokens that appear frequently among high-reward candidates, while the factorized policy preserves enough exploration to escape local optima.

Theoretical guarantee. Write $f(x) = f(x; I, T_{\text{ext}})$ for the contrastive reward and assume that it decomposes additively across positions $f(x) = \sum_{l=1}^L f_l(x[l])$ for $x \in \mathcal{V}^L$ (we discuss in Appendix A.1 why a first-order surrogate is locally additive). The following theorem formalizes the token-amplification effect underlying RGPO.

Theorem 3.1 (RGPO (ε, δ) -guarantee) *Fix any $\varepsilon > 0$. If the number of RGPO iterations satisfies $n = \Omega(\log |\mathcal{V}|)$ and the number of candidates at each iteration satisfies $B = \Omega(\log(1/\delta))$, then $\hat{T}_{\text{ret}}$ from the final policy satisfies*

$$f(\hat{T}_{\text{ret}}) \geq \max_{x \in \mathcal{V}^L} f(x) - \varepsilon$$

with probability at least $1 - \delta$. See proof in Appendix A.1.

3.3 Extraction Segment Design

The retrieval segment determines whether the test image enters the generator’s context. The remaining question is how to design the extraction segment T_{ext} and the scoring rule so that, together, they elicit discriminative MIA signals: low scores for non-members and high scores for members. We therefore consider a query $T = T_{\text{ext}} \oplus T_{\text{ret}}$ and analyze the MIA score induced by a *general* scoring rule.

Let $\mathcal{X}$ denote the set of retrieved images, and let $Y \sim p(y \mid \mathcal{X}, T)$ be the generator output. We define a measurable score function $s : \mathcal{I} \times \mathcal{Y} \rightarrow \mathbb{R}$, where the first argument is the test image and the second is the generated output. The single-query MIA score is the random variable $S = s(I, Y)$. Let $Z \in \{0, 1\}$ indicate whether the test image I belongs to the target database. For $z \in \{0, 1\}$, define the conditional score distribution under membership status z as

$$F_z(a \mid I, T) = \Pr(S \leq a \mid Z = z, I, T).$$

We further define the retrieval hit event $H = \{I \in \mathcal{X}\}$, i.e., the event that the test image is retrieved. Let $\rho_T(I) = \Pr(H = 1 \mid Z = 1, I, T)$ denote the retrieval probability induced by the query T . We also define the hit and miss score distributions as

$$F_{\text{hit}}(a \mid I, T) = \Pr(S \leq a \mid H = 1, I, T), \quad F_{\text{miss}}(a \mid I, T) = \Pr(S \leq a \mid H = 0, I, T).$$

Since a non-member image cannot be retrieved from the target database, we have $H = 0$ whenever $Z = 0$. Hence, the member and non-member score distributions satisfy the mixture identity

$$F_1(a \mid I, T) = \rho_T(I) F_{\text{hit}}(a \mid I, T) + (1 - \rho_T(I)) F_{\text{miss}}(a \mid I, T), \quad F_0(a \mid I, T) = F_{\text{miss}}(a \mid I, T).$$

This identity separates the two roles of our attack: the retrieval segment optimized by RGPO increases the retrieval probability $\rho_T(I)$, while the extraction segment and the scoring function determine the separation between F_{hit} and F_{miss} . Thus, a successful attack requires both a large retrieval probability and a scoring rule whose hit/miss gap is large. Equivalently, the extraction segment should make outputs in the hit case systematically more similar to the test image than outputs in the miss case.

Guided by this principle, we instantiate the extraction segment as a prompt that asks the generator to describe the retrieved reference images. We then define the MIA score as the similarity between the

test image I and the generated output Y . Because T2I and Q&A systems expose different interaction patterns, we employ task-specific adaptations of this extraction-and-scoring strategy, described below.

(a) T2I. Since T2I generators [26] lack instruction tuning, prompts such as “*please describe the retrieved images*” are largely ignored (low text attention) and easily flagged as anomalous requests (as illustrated in Appendix B). To address this, we use the caption of the test image as the extraction segment, leveraging the copy-like behavior of generators under aligned inputs. If I is successfully retrieved as a reference (i.e., a member), the T2I generator faithfully reproduces its content. Even when the retrieved image is not I , the generator still tends to reproduce salient content from the reference, as it shares similar semantics with the caption. Together, these effects make caption-based extraction an effective realization of our design principle. As illustrated in Appendix F, members trigger near-verbatim reconstruction through the copy-like behavior, while non-members yield only semantically aligned but visually divergent outputs. **(b) Q&A.** A direct command such as “*describe each retrieved image*” causes the VLM to enumerate all retrieved images in a single response, introducing substantial noise into the similarity computation. To address this, we split the output into per-image descriptions and define the MIA score as the maximum similarity between I and any individual description. As shown in Table 3, we further observe that computing similarity within the same modality (i.e., T2T or I2I) is essential for achieving strong performance.

3.4 The Overall Pipeline and Multi-query Aggregation

Our single-query attack proceeds in four steps: (1) construct an extraction segment T_{ext} following Sec. 3.3; (2) construct a retrieval segment T_{ret} following Sec. 3.2; (3) feed the composite query $T = T_{\text{ext}} \oplus T_{\text{ret}}$ into the target system and obtain its output; and (4) compute the MIA score: image-to-image similarity between the test image and the output in the T2I setting, or the maximum text-to-text similarity between the test image description and each output fragment in the Q&A setting.

Multi-query aggregation. Some queries may fail to elicit a meaningful MIA signal due to errors in optimizing the retrieval segment or approximating the appropriate extraction segment. To improve reliability, we issue $M \times N$ queries by pairing each of M extraction segments with N optimized retrieval segments. We then apply K-means clustering to the resulting scores, retain only the high-scoring cluster, and aggregate the retained scores to produce the final MIA decision.

4 Experiments

We aim to answer the following research questions: (1) How does ImageAuditor perform compared to adapted baselines under both T2I and Q&A settings? (Sec. 4.2) (2) Is ImageAuditor robust under various settings? (Sec. 4.3) (3) How do our attack designs contribute to ImageAuditor? (Sec. 4.3)

4.1 Experimental Setup

Image database and evaluation datasets. For both T2I and Q&A tasks, we construct large-scale image databases following prior work [22, 6]. **(1) T2I tasks.** We construct a database of 259,057 images spanning six datasets: MSCOCO [17], ImageNet-100 subset [30], WikiArt [25], Stanford Cars [15], Stanford Dogs [7], and CelebA-HQ [13]. We use each dataset’s training split to construct the database, and sample member and non-member images from the training and test splits. Due to computational constraints, we randomly select three of the six datasets for attack evaluation. Details can be found in Appendix D.1. For utility evaluation, we follow the standard protocol [22] as illustrated in Appendix D.1. **(2) Q&A tasks.** Following [6], we use MMQA [34] and the MSCOCO training split as the database, containing 58,075 and 118,287 images, respectively. Since MMQA does not provide a training/test split, we randomly hold out 1,000 images as non-members and treat the rest as members. For utility evaluation, we follow the same setup as AQUA [3].

Models. For T2I tasks, we employ four generators: SDXL [26] plus IP-Adapter [40] (default), SD1.5 [29] plus IP-Adapter, Kandinsky v2.2 [28], and SDXL plus Conceptrol [11], paired with CLIP-ViT-H/14 [27] as the retriever. For Q&A tasks, we employ Qwen2.5-VL-7B-Instruct [37] (default) and LLaVA-1.6-Mistral-7B [19] as generators, with CLIP-ViT-L/14-336 as the retriever. The setup follows standard practice in the IRAG literature [22, 6].

Metrics. We report AUROC, ACC, TPR@5%FPR, and Retrieval Success Rate (RSR) for attack evaluation. Details of the attack and utility evaluation metrics are provided in Appendix D.3.

Table 1: Attack results of ImageAuditor and adapted baselines* on IRAG for T2I tasks. For adapted baselines, we use the same extraction segments and optimize retrieval segments using their respective methods. The system uses CLIP ViT-H/14 for retrieval and SDXL+IP-Adapter for generation. $\text{TPR}_{5\%}$ denotes $\text{TPR}@5\%\text{FPR}$. Detailed TPR-FPR curves are provided in Appendix Figure 7.

Attack Method	MSCOCO				WikiArt				Stanford Dogs			
	AUC	ACC	$\text{TPR}_{5\%}$	RSR	AUC	ACC	$\text{TPR}_{5\%}$	RSR	AUC	ACC	$\text{TPR}_{5\%}$	RSR
Naïve	0.72	0.70	0.37	0.53	0.63	0.63	0.31	0.33	0.66	0.67	0.29	0.40
PoisonedRAG-B*	0.71	0.71	0.26	0.55	0.68	0.66	0.33	0.42	0.70	0.68	0.38	0.48
PoisonedRAG*	0.79	0.77	0.45	0.66	0.69	0.70	0.40	0.53	0.72	0.74	0.40	0.52
BadRAG*	0.80	0.78	0.46	0.70	0.66	0.67	0.30	0.48	0.71	0.71	0.33	0.54
ImageAuditor	0.91	0.85	0.55	0.96	0.87	0.82	0.49	0.93	0.90	0.84	0.65	0.90

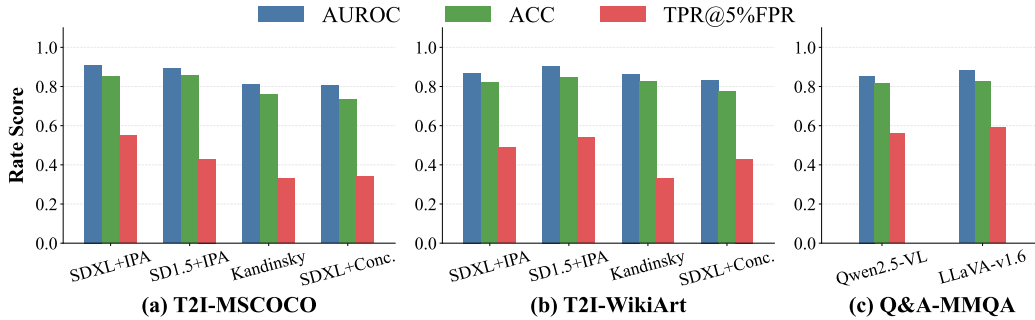


Figure 3: ImageAuditor remains effective across different generators for both T2I and Q&A tasks.

Baselines. We compare ImageAuditor against four baselines. *Naïve* directly uses the caption of the test image as the query, with no retrieval segment. *PoisonedRAG-B**, *PoisonedRAG** and *BadRAG** are adapted baselines that retain our extraction segment but optimize the retrieval segment using techniques from existing RAG poisoning attacks [43, 39] (See Appendix D.4). We do not include previous text-RAG MIAs, as their Q&A-based pipelines do not transfer to IRAG. We also exclude jailbreaking methods like GCG [42], as they are less effective than PoisonedRAG in this setting [43].

Evaluation setup. Unlike prior work, our image database contains samples drawn from multiple sources (e.g., MSCOCO, WikiArt). In practice, an auditor is typically interested in a specific data domain (e.g., artwork) and collects calibration samples from that domain to set the threshold and infer membership. We therefore report per-dataset results, which better simulate this realistic auditing scenario, and defer the combined attack setting to Appendix Table 6.

(1) *Attack settings.* The attacker is allowed $M \times N$ queries per test image, where M is the number of extraction segments and N is the number of retrieval segments optimized per extraction segment; we set $M = N = 2$ by default. For the extraction segment, we use Qwen-VL to generate and rank candidate captions in the T2I setting, and manually design structured prompts in the Q&A setting. For the retrieval segment, we set the token length to 8 and sample a small shadow set of 32 images (disjoint from the database). By default, the attacker has access to E_{tgt} ; under the restrictive setting, we instead assume access to shadow embedding models, which are smaller and trained with different data distributions [38, 33]. The scoring model is the target (or a selected shadow) embedding model for T2I and a text-embedding model E5 [36] for Q&A. Due to the high experimental cost (~ 1 min/sample), we use a 200 member/non-member set per subdataset and verify the stability in Appendix Table 5. Remaining hyperparameters (optimization iterations, RGPO temperature, etc.) are deferred to Appendix D.5. (2) *IRAG settings.* The retriever uses $K = 1$ for T2I and $K = 2$ for Q&A by default, following the literature [40, 3]; this small retrieval budget poses a substantial challenge for our attack. Other generation settings follow each generator’s default setup (Appendix D.2).

4.2 Main Results

In this part, we discuss how different MIAs perform against IRAG. We further validate that IRAG substantially improves utility on both T2I and Q&A tasks, and defer this analysis to Appendix C.

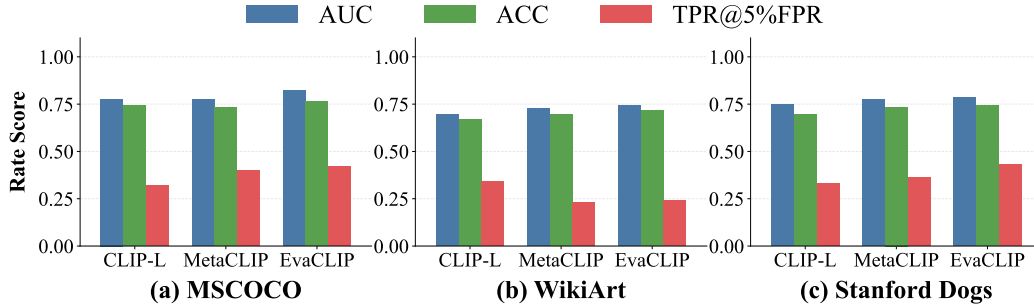


Figure 4: ImageAuditor remains effective under strict black-box settings. We use a weaker shadow embedding model (e.g., with different architectures or training distributions) to optimize the retrieval segments and compute the MIA scores. The target system uses CLIP ViT-H/14 for retrieval.

Table 2: Fine-grained captions for the extraction segment elicit stronger copy-like behavior and stronger attacks than generic captions (Gen.).

Type	MSCOCO			WikiArt		
	AUC	ACC	TPR _{5%}	AUC	ACC	TPR _{5%}
Gen.	0.71	0.70	0.38	0.75	0.72	0.39
Fine.	0.91	0.85	0.55	0.87	0.82	0.49

Table 3: ImageAuditor achieves stronger attacks when scoring within a single modality (Single) than across modalities (Cross).

Type	T2I-MSCOCO			Q&A-MMQA		
	AUC	ACC	TPR _{5%}	AUC	ACC	TPR _{5%}
Cross	0.60	0.59	0.14	0.82	0.79	0.37
Single	0.91	0.85	0.55	0.85	0.82	0.56

ImageAuditor outperforms baseline attacks on IRAG systems for T2I and Q&A tasks. We present the main results of ImageAuditor and adapted baselines on IRAG for T2I tasks in Table 1 and for Q&A tasks in Appendix Table 4. All methods share the same extraction segment and differ only in retrieval segment optimization. (1) Our attack achieves state-of-the-art performance on both T2I and Q&A tasks, outperforming existing attacks by a large margin (e.g., over 10% AUROC on T2I tasks). Existing attacks suffer from low target retrieval rates because they easily get trapped in local minima within the cross-modal embedding space. (2) Our attack is effective in the low false positive region (also illustrated in Figure 7 and 8), achieving around 50% TPR@5%FPR, which is commonly interpreted as a strong attack in the literature [1]. (3) In the T2I setting, WikiArt shows lower performance than other datasets because diffusion models exhibit weaker copy-like behavior on artwork references, as stylized images are less aligned with the text. (4) In the Q&A setting, our attack is more effective on MMQA due to the uniqueness of its images (e.g., film posters). (5) Existing baselines perform poorly in the Q&A setting because when the extraction segment (“Describe each retrieved...”) is unrelated to the test image, the initial query embedding lies far from the target, making gradient-based optimization challenging—a regime where RGPO’s reward-guided updates succeed.

ImageAuditor remains effective across diverse generators for both T2I and Q&A tasks. For T2I, beyond the default SDXL+IPA, we evaluate generators varying along two axes—diffusion backbone and conditioning mechanism—including SD1.5+IPA (different backbone), Kandinsky (different family, adapter-free), and SDXL+Conceptrol (alternative conditioning). As shown in Figure 3(a,b), ImageAuditor maintains high AUROC (>0.80) across all variants. Conceptrol yields the weakest attack due to its more creative generation capability, yet it still exceeds 0.30 TPR@5%FPR. For Q&A, Qwen2.5-VL and LLaVA-v1.6 produce nearly identical results (Figure 3(c)), as both VLMs faithfully describe image content. Additional results are provided in the Appendix Figure 9. Overall, these results confirm that ImageAuditor is robust to the choice of underlying generators.

4.3 Ablation Study

Unless otherwise mentioned, we conduct ablation studies on IRAG for T2I tasks.

ImageAuditor remains effective in the restrictive attack setting. Under restrictive constraints, where the attacker has no access to the embedding model used by the target system, we construct retrieval segments using a weaker shadow embedding model. Despite this mismatch, Figure 4 shows that ImageAuditor still achieves an AUROC of around 0.75, which is generally considered strong for

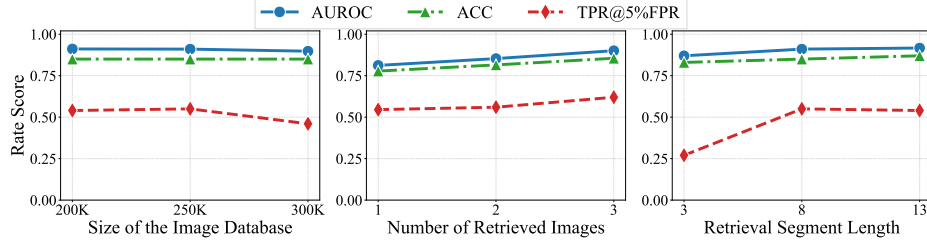


Figure 5: Impact of (a) the size of the image database, and (b) the number of retrieved images, and (c) the retrieval segment length on attack results. ImageAuditor remains robust across all these settings.

MIA. The observed performance drop stems not only from weaker retrieval segment construction but also from the weaker scoring model used in this setup; as shown in Appendix Table 11, stronger scoring models consistently yield better results by capturing more fine-grained image semantics. Interestingly, EvaCLIP transfers best on MSCOCO and Dogs but degrades on WikiArt, likely due to lower embedding similarity with the target (ViT-H/14) on artwork images.

Our tailored attack designs lead to stronger attack results. We further investigate the impact of our attack designs for the *extraction segment*, *retrieval segment*, and *scoring mechanism*. For the extraction segment, as shown in Table 2 and Appendix Table 9, fine-grained captions generated by BLIP and Qwen-VL significantly improve the attack results, yielding up to $\sim 10\%$ gains in AUROC. We attribute this to the fact that high-quality captions more effectively elicit copy-like behavior from diffusion models and provide a better starting point for retrieval segment construction. For the retrieval segment, we have demonstrated the superiority of the proposed RGPO mechanism in Table 1. In Appendix Table 10, we further evaluate the impact of incorporating a small shadow set for contrastive reward. Under the restrictive setting, we show that negative samples consistently improve attack performance by modeling a more favorable optimization landscape that generalizes. Beyond these two components, we also examine different strategies for computing the similarity score. In both T2I and Q&A settings, similarity can be measured between the output and the image (image modality) or between the output and image descriptions (text modality). As shown in Table 3, we observe that computing similarity within the same modality yields large improvements, especially for $TPR_{5\%}$; accordingly, we extract image details from the test image in the Q&A setting.

ImageAuditor remains effective under different IRAG setups. In Figure 5, we investigate the impact of the size of the image database, the number of retrieved images, and the length of the retrieval segment on attack results. In (a), we enlarge the image database by incorporating additional images from ImageNet and observe that the database size has only a minor effect on attack performance. This is because RGPO can effectively retrieve the test image whenever it lies in the database. We further observe that a larger retrieval budget slightly improves the attack in the Q&A setting, as $K=3$ increases the likelihood of the members being retrieved. Here, we experiment with Q&A setting due to architectural constraints in T2I generators. Finally, ImageAuditor saturates even when the retrieval segment length is small (8 tokens), which makes our attack stealthier and less likely to be detected.

Ablation on Multi-Query Attack Strategies. We evaluate the impact of aggregation strategy and query number for multi-query attacks in Appendix Figures 10 and 11. We find that the K-means-based strategy balances precision and coverage, yielding more stable attack performance across settings. We also observe that increasing the number of queries improves attack results, though notably, even a single-query attack achieves competitive performance. Detailed analysis is provided in Appendix C.

5 Conclusion

We introduce *ImageAuditor*, the first membership inference attack (MIA) against image-based RAG (IRAG) systems. To overcome fundamental limitations of prior text-RAG MIAs, *ImageAuditor* decomposes each attack query into a retrieval segment and an extraction segment. We propose Reward-Guided Policy Optimization to enhance cross-modal retrieval, and design tailored prompting strategies paired with a scoring rule to elicit discriminative MIA signals. Extensive experiments show that *ImageAuditor* achieves over 80% AUROC with only four queries and remains effective under restrictive attack settings. These results establish ImageAuditor as a practical auditing tool.

Limitations and Future Work. This work focuses on establishing the feasibility of MIAs against IRAG. While our attack achieves strong performance across multiple settings, we evaluate robustness only against text/image attention ratio-based detection (see Appendix B); more sophisticated defenses, such as differential privacy [14] and output perturbation, remain unexplored. In addition, we focus on text-based retrieval and do not evaluate multimodal inputs. However, as discussed in Appendix E, our attack naturally extends to this setting. We leave these directions to future work.

References

- [1] Nicholas Carlini, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramer. Membership inference attacks from first principles. In *2022 IEEE symposium on security and privacy (SP)*, pages 1897–1914. IEEE, 2022.
- [2] Harsh Chaudhari, Giorgio Severi, John Abascal, Matthew Jagielski, Christopher A Choquette-Choo, Milad Nasr, Cristina Nita-Rotaru, and Alina Oprea. Phantom: General trigger attacks on retrieval augmented language generation. 2024.
- [3] Tianyu Chen, Jian Lou, and Wenjie Wang. Safeguarding multimodal knowledge copyright in the rag-as-a-service environment. *arXiv preprint arXiv:2506.10030*, 2025.
- [4] Wenhui Chen, Hexiang Hu, Xi Chen, Pat Verga, and William Cohen. Murag: Multimodal retrieval-augmented generator for open question answering over images and text. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 5558–5570, 2022.
- [5] Wenhui Chen, Hexiang Hu, Chitwan Saharia, and William W Cohen. Re-imagen: Retrieval-augmented text-to-image generator. *arXiv preprint arXiv:2209.14491*, 2022.
- [6] Zhanpeng Chen, Chengjin Xu, Yiyang Qi, and Jian Guo. Mllm is a strong reranker: Advancing multimodal retrieval-augmented generation via knowledge-enhanced reranking and noise-injected training. *arXiv preprint arXiv:2407.21439*, 2024.
- [7] E Dataset. Novel datasets for fine-grained image categorization. In *First workshop on fine grained visual categorization, CVPR. Citeseer. Citeseer. Citeseer*, volume 5, page 2. Citeseer, 2011.
- [8] Mingkai Deng, Jianyu Wang, Cheng-Ping Hsieh, Yihan Wang, Han Guo, Tianmin Shu, Meng Song, Eric Xing, and Zhiting Hu. Rlprompt: Optimizing discrete text prompts with reinforcement learning. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3369–3391, 2022.
- [9] Javid Ebrahimi, Anyi Rao, Daniel Lowd, and Dejing Dou. Hotflip: White-box adversarial examples for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 31–36, 2018.
- [10] Runpeng Geng, Yanting Wang, Ying Chen, and Jinyuan Jia. Unic-rag: Universal knowledge corruption attacks to retrieval-augmented generation. *arXiv preprint arXiv:2508.18652*, 2025.
- [11] Qiyuan He and Angela Yao. Conceptrol: Concept control of zero-shot personalized image generation. *arXiv preprint arXiv:2503.06568*, 2025.
- [12] Yu He, Boheng Li, Liu Liu, Zhongjie Ba, Wei Dong, Yiming Li, Zhan Qin, Kui Ren, and Chun Chen. Towards {Label-Only} membership inference attack against pre-trained large language models. In *34th USENIX Security Symposium (USENIX Security 25)*, pages 1609–1628, 2025.
- [13] Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. Progressive growing of gans for improved quality, stability, and variation. *arXiv preprint arXiv:1710.10196*, 2017.
- [14] Tatsuki Koga, Ruihan Wu, Zhiyuan Zhang, and Kamalika Chaudhuri. Privacy-preserving retrieval-augmented generation with differential privacy. *arXiv preprint arXiv:2412.04697*, 2024.

- [15] Jonathan Krause, Michael Stark, Jia Deng, and Li Fei-Fei. 3d object representations for fine-grained categorization. In *Proceedings of the IEEE international conference on computer vision workshops*, pages 554–561, 2013.
- [16] Yuying Li, Gaoyang Liu, Chen Wang, and Yang Yang. Generating is believing: Membership inference attacks against retrieval-augmented generation. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE, 2025.
- [17] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *European conference on computer vision*, pages 740–755. Springer, 2014.
- [18] Weizhe Lin and Bill Byrne. Retrieval augmented visual question answering with outside knowledge. In *Proceedings of the 2022 conference on empirical methods in natural language processing*, pages 11238–11254, 2022.
- [19] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- [20] Mingrui Liu, Sixiao Zhang, and Cheng Long. Mask-based membership inference attacks for retrieval-augmented generation. In *Proceedings of the ACM on Web Conference 2025*, pages 2894–2907, 2025.
- [21] Ziyuan Luo, Yangyi Zhao, Ka Chun Cheung, Simon See, and Renjie Wan. Imagesentinel: Protecting visual datasets from unauthorized retrieval-augmented image generation. *arXiv preprint arXiv:2510.12119*, 2025.
- [22] Yuanhuiyi Lyu, Xu Zheng, Luta Jiang, Yibo Yan, Xin Zou, Huiyu Zhou, Linfeng Zhang, and Xuming Hu. Realrag: Retrieval-augmented realistic image generation via self-reflective contrastive learning. In *International Conference on Machine Learning*, pages 41772–41790. PMLR, 2025.
- [23] Ali Naseh, Yuefeng Peng, Anshuman Suri, Harsh Chaudhari, Alina Oprea, and Amir Houmansadr. Riddle me this! stealthy membership inference for retrieval-augmented generation. In *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, pages 1245–1259, 2025.
- [24] Yan Pang and Tianhao Wang. Black-box membership inference attacks against fine-tuned diffusion models. *arXiv preprint arXiv:2312.08207*, 2023.
- [25] Fred Phillips and Brandy Mackintosh. Wiki art gallery, inc.: A case for critical thinking. *Issues in Accounting Education*, 26(3):593–608, 2011.
- [26] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. Sdxl: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952*, 2023.
- [27] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.
- [28] Anton Razhigaev, Arseniy Shakhmatov, Anastasia Maltseva, Vladimir Arkhipkin, Igor Pavlov, Ilya Ryabov, Angelina Kuts, Alexander Panchenko, Andrey Kuznetsov, and Denis Dimitrov. Kandinsky: an improved text-to-image synthesis with image prior and latent diffusion. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 286–295, 2023.
- [29] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.

- [30] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. Imagenet large scale visual recognition challenge. *International journal of computer vision*, 115(3):211–252, 2015.
- [31] Rotem Shalev-Arkushin, Rinon Gal, Amit H Bermano, and Ohad Fried. Imagerag: Dynamic image retrieval for reference-guided image generation. *arXiv preprint arXiv:2502.09411*, 2025.
- [32] Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE symposium on security and privacy (SP)*, pages 3–18. IEEE, 2017.
- [33] Quan Sun, Yuxin Fang, Ledell Wu, Xinlong Wang, and Yue Cao. Eva-clip: Improved training techniques for clip at scale. *arXiv preprint arXiv:2303.15389*, 2023.
- [34] Alon Talmor, Ori Yoran, Amnon Catav, Dan Lahav, Yizhong Wang, Akari Asai, Gabriel Ilharco, Hannaneh Hajishirzi, and Jonathan Berant. Multimodalqa: Complex question answering over text, tables and images. *arXiv preprint arXiv:2104.06039*, 2021.
- [35] Haowei Wang, Rupeng Zhang, Junjie Wang, Mingyang Li, Yuekai Huang, Dandan Wang, and Qing Wang. Joint-gcg: Unified gradient-based poisoning attacks on retrieval-augmented generation systems. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 35793–35801, 2026.
- [36] Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*, 2022.
- [37] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.
- [38] Hu Xu, Saining Xie, Xiaoqing Ellen Tan, Po-Yao Huang, Russell Howes, Vasu Sharma, Shang-Wen Li, Gargi Ghosh, Luke Zettlemoyer, and Christoph Feichtenhofer. Demystifying clip data. *arXiv preprint arXiv:2309.16671*, 2023.
- [39] Jiaqi Xue, Mengxin Zheng, Yebowen Hu, Fei Liu, Xun Chen, and Qian Lou. Badrag: Identifying vulnerabilities in retrieval augmented generation of large language models. *arXiv preprint arXiv:2406.00083*, 2024.
- [40] Hu Ye, Jun Zhang, Sibio Liu, Xiao Han, and Wei Yang. Ip-adapter: Text compatible image prompt adapter for text-to-image diffusion models. *arXiv preprint arXiv:2308.06721*, 2023.
- [41] Taeyoung Yun, Dinghuai Zhang, Jinkyoo Park, and Ling Pan. Learning to sample effective and diverse prompts for text-to-image generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 23625–23635, 2025.
- [42] Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.
- [43] Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. {PoisonedRAG}: Knowledge corruption attacks to {Retrieval-Augmented} generation of large language models. In *34th USENIX Security Symposium (USENIX Security 25)*, pages 3827–3844, 2025.

A Proofs

A.1 Proof of Theorem 3.1

We prove the theorem for $\beta = 1$. $\beta < 1$ introduces additional approximation error but does not change the qualitative amplification mechanism.

For each position $l \in \{1, \dots, L\}$, define

$$\mathcal{G}_l := \left\{ v \in \mathcal{V} : f_l(v) \geq \max_{v \in \mathcal{V}} f_l(v) - \frac{\varepsilon}{L} \right\}.$$

If $\mathcal{G}_l = \mathcal{V}$, then every token at position l is already ε/L -good and there is nothing to prove for that position. Otherwise define the good-bad margin $\Delta_l := \min_{v \in \mathcal{G}_l} f_l(v) - \max_{v \notin \mathcal{G}_l} f_l(v) > 0$ and $\Delta := \min_l \Delta_l > 0$.

We note that any sequence whose token at each position lies in the corresponding good set is ε -optimal. To see that, for any $x \in \mathcal{V}^L$ with $x[l] \in \mathcal{G}_l$ for all l , we have

$$\max_{z \in \mathcal{V}^L} f(z) - f(x) = \sum_{l=1}^L \left(\max_{v \in \mathcal{V}} f_l(v) - f_l(x[l]) \right) \leq \sum_{l=1}^L \frac{\varepsilon}{L} = \varepsilon.$$

Therefore, it suffices to show that the modal token of the final policy belongs to $\mathcal{G}_l$ at every position.

To see the token-amplification effect, we first analyze the one-step amplification. Fix an iteration t and a position l . Let the current product policy be $\pi_t = \prod_{j=1}^L \pi_{t,j}$, and let $X \sim \pi_t$. Define the importance weight $w_t(x) := \exp(f(x)/\tau) \mathbf{1}\{f(x) \geq \gamma_t\}$. For each token $v \in \mathcal{V}$, let $m_{t,l}(v) := \mathbb{E}[w_t(X) \mid X[l] = v, \pi_t]$. Since the reward is additive, if we write $S_{t,l} := \sum_{j \neq l} f_j(X[j])$, then we have

$$m_{t,l}(v) = \exp\left(\frac{f_l(v)}{\tau}\right) \mathbb{E}\left[\exp\left(\frac{S_{t,l}}{\tau}\right) \mathbf{1}\{S_{t,l} \geq \gamma_t - f_l(v)\} \mid \pi_t\right].$$

For any $v \in \mathcal{G}_l$ and $u \notin \mathcal{G}_l$, since $f_l(v) - f_l(u) \geq \Delta_l \geq \Delta$, we have

$$\begin{aligned} m_{t,l}(v) &\geq \exp\left(\frac{f_l(v) - f_l(u)}{\tau}\right) \exp\left(\frac{f_l(u)}{\tau}\right) \mathbb{E}\left[\exp\left(\frac{S_{t,l}}{\tau}\right) \mathbf{1}\{S_{t,l} \geq \gamma_t^* - f_l(u)\} \mid \pi_t\right] \\ &\geq \exp\left(\frac{\Delta}{\tau}\right) m_{t,l}(u). \end{aligned}$$

Define the exact marginal update induced by w_t

$$p_{t+1,l}(v) = \frac{\mathbb{E}[w_t(X) \mathbf{1}\{X[l] = v\} \mid \pi_t]}{\mathbb{E}[w_t(X) \mid \pi_t]} = \frac{\pi_{t,l}(v) m_{t,l}(v)}{\sum_{u \in \mathcal{V}} \pi_{t,l}(u) m_{t,l}(u)},$$

and define

$$q_{t,l} := \sum_{v \in \mathcal{G}_l} \pi_{t,l}(v), \quad \bar{q}_{t+1,l} := \sum_{v \in \mathcal{G}_l} p_{t+1,l}(v).$$

Then we have

$$\bar{q}_{t+1,l} = \frac{\sum_{v \in \mathcal{G}_l} \pi_{t,l}(v) m_{t,l}(v)}{\sum_{v \in \mathcal{G}_l} \pi_{t,l}(v) m_{t,l}(v) + \sum_{v \notin \mathcal{G}_l} \pi_{t,l}(v) m_{t,l}(v)} \geq \frac{\kappa q_{t,l}}{(1 - q_{t,l}) + \kappa q_{t,l}},$$

where $\kappa = \exp(\Delta/\tau) > 1$. Equivalently,

$$\frac{\bar{q}_{t+1,l}}{1 - \bar{q}_{t+1,l}} \geq \kappa \frac{q_{t,l}}{1 - q_{t,l}}.$$

Therefore the update multiplies the good-token odds by at least the factor κ at every iteration.

Given the amplification result, we can now bound the number of RGPO iterations n and the number of samples at each iteration B . Since RGPO is initialized uniformly, for every position l we have $q_{1,l} \geq 1/|\mathcal{V}|$. The amplification result then implies that, after $n = \Omega(\log |\mathcal{V}|)$ iterations, the good-token mass at each position satisfies $q_{n,l} \geq 1 - 1/(2L)$. Therefore, for a single sample

$X \sim \pi_n$ from the final policy, a union bound over the L positions gives $\Pr(X[l] \in \mathcal{G}_l \text{ for all } l) \geq 1 - \sum_{l=1}^L \Pr(X[l] \notin \mathcal{G}_l) \geq 1 - L/(2L) = 1/2$. Whenever this event occurs, the additive structure of f implies $f(X) \geq \max_{x \in \mathcal{V}^L} f(x) - \varepsilon$. Since each draw is ε -optimal with probability at least $1/2$, by the independent sampling from the policy, choosing $B = \Omega(\log(1/\delta))$ ensures that, with probability at least $1 - \delta$,

$$f(\hat{T}_{\text{ret}}) \geq \max_{x \in \mathcal{V}^L} f(x) - \varepsilon.$$

This completes the proof.

On the additivity assumption. The cosine reward is not globally position-additive. However, for a fixed extraction segment T_{ext} and a short retrieval suffix, a first-order Taylor expansion of $\text{sim}(E(T_{\text{ext}} \oplus T_{\text{ret}}), E(I))$ around the current policy mean is position-separable:

$$f(x) \approx c + \sum_{l=1}^L \langle g_l, \phi(x[l]) \rangle,$$

where g_l is the gradient with respect to the l -th token embedding and ϕ is the embedding lookup. Theorem 3.1 should therefore be read as a guarantee on this local surrogate, which RGPO effectively optimizes through reward-ranked sampling. The empirical retrieval success rates (90%+) in our experiments (Table 1) are consistent with this approximation being tight in the regimes we evaluate.

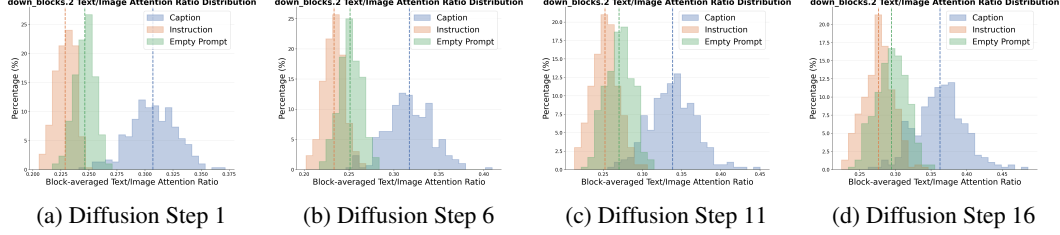


Figure 6: Instruction-based extraction segments produce abnormal text-image attention ratios due to the lack of semantic meaning in text prompts, which can be easily detected in early diffusion steps.

B Text/Image Attention Ratio based Detection

In the T2I setting, a naïve attacker may use an instruction prompt (e.g., “Please describe the retrieved images”) or even an empty prompt (e.g., “”) to coerce the T2I generator into directly outputting the retrieved image. The underlying intuition is that, in such cases, the text input is largely ignored and generation is governed primarily by the image reference, leading to direct leakage of the retrieved image. While effective in certain cases, we show that this strategy can be easily detected by a simple *Text/Image Attention Ratio* mechanism.

Detection mechanism. Let’s take SDXL+IP-Adapter as an example. The defender collects cross-attention scores from a specific U-Net block (we use `down_blocks.2`) of the diffusion model. In the IP-Adapter architecture, each cross-attention layer contains two parallel branches: one attending to text embeddings and one attending to image embeddings. Let $A_{\text{text}}^{(l)}$ and $A_{\text{img}}^{(l)}$ denote the attention scores from the text and image branches at layer l , respectively. The text/image attention ratio A is computed as:

$$A = \frac{1}{L} \sum_{l=1}^L \frac{\|A_{\text{text}}^{(l)}\|_1}{\|A_{\text{img}}^{(l)}\|_1 + \epsilon}, \quad (1)$$

where L is the number of cross-attention layers in that block and ϵ is a small constant for numerical stability. A low ratio indicates that the generation is dominated by image conditioning, signaling a potential leakage attack.

Detection results. As shown in Figure 6, both instruction and empty prompts produce abnormally low text/image attention ratios at early diffusion steps, making them easily distinguishable from benign queries. In contrast, our caption-based attack uses natural text queries and yields ratios within the normal range, thereby evading detection. Moreover, while the empty prompt provides a strong shortcut for image leakage, it underperforms our caption-based approach in eliciting reference images, achieving a CLIP image similarity of 0.815 compared to 0.833 for ours (note that we directly compare the similarity between the reference image and the output image, rather than measuring MIA performance). The empty prompt also complicates retrieval segment construction, as the absence of textual context weakens the optimization signal. Together, these results demonstrate that our extraction segment design is both *stealthy* and *effective*.

C Additional Experiment Analysis

IRAG improves the utility of both T2I and Q&A tasks. As shown in Appendix Tables 7 and 8, IRAG substantially improves T2I generation utility (measured by CLIP-I) by over 10% on the ImageNet100, WikiArt, Dogs, and Cars datasets, and boosts response accuracy on MMQA from 30% to approximately 55%. These gains stem from the inherent limitations of parametric knowledge in generative models: (1) fine-grained visual concepts, long-tail entities, and domain-specific details are often poorly learned and underrepresented in the model’s parameters; and (2) novel concepts that emerge after the base diffusion model or VLM is trained remain entirely absent from its parametric knowledge. Both cases lead to hallucinated or generic outputs. By retrieving semantically relevant images at inference time, IRAG grounds generation in concrete visual evidence, allowing the model to faithfully reproduce category-specific details in T2I tasks and to answer visually grounded questions that would otherwise exceed its parametric capacity. This confirms that IRAG serves as an effective

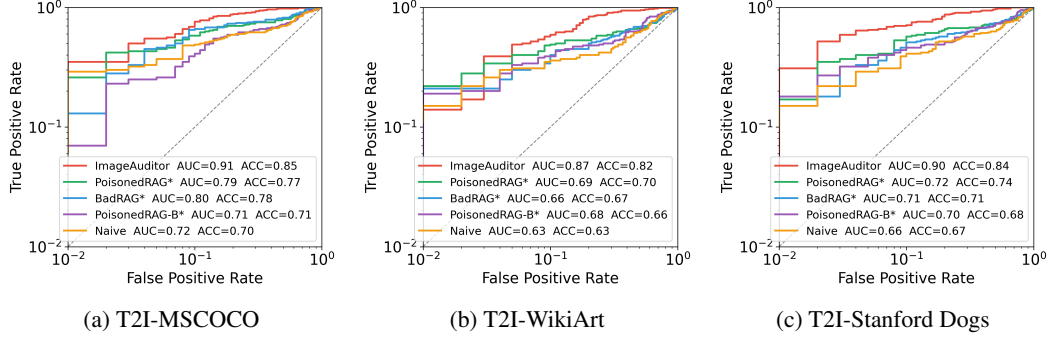


Figure 7: TPR-FPR curves of ImageAuditor and adapted baselines on IRAG for T2I generation tasks.

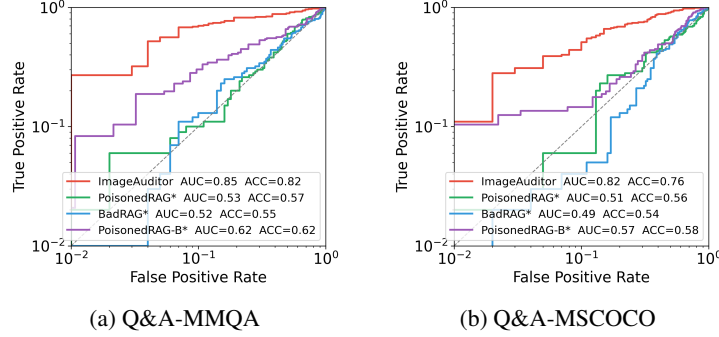


Figure 8: TPR-FPR curves of ImageAuditor and adapted baselines on IRAG for Q&A tasks. Adapted PoisonedRAG and BadRAG fail because the extraction segment—“Describe each retrieved image. Format as ‘Image 1:’, ‘Image 2:’, etc.”—provides no semantic alignment with the target image, causing prior attacks to get trapped in local minima and fail to retrieve that image.

external memory, complementing the model’s internal knowledge with up-to-date and fine-grained visual information.

Detailed Analysis of Multi-Query Attack Strategies. We evaluate different variants of attack strategies for multi-query attacks in Appendix Figures 10 and 11. Specifically, we observe that different aggregation strategies achieve comparable performance under the standard setting, but the average-score-based approach exhibits much lower performance ($\sim 10\%$ drop in TPR@5%FPR) under the restrictive setting. We explain that under this more challenging setting, many queries fail to retrieve the test image despite its presence in the database and, therefore, produce noisy scores. The K-means-based strategy mitigates this by aggregating scores from high-confidence clusters, which balances precision with coverage and yields more stable performance across settings. In Figure 11, we show that increasing the number of queries slightly improves attack results, while notably, even the single-query attack achieves desirable performance.

D Additional Details

D.1 Description of the Datasets.

For both T2I and Q&A tasks, we construct large-scale image databases by aggregating data from multiple sources, following prior work [22, 6]. For datasets without official splits, we partition the data into member and non-member sets, using the member split to build the image database.

T2I datasets. (1) *MSCOCO (2017)* [17] is a large-scale image-caption dataset of real-world photographic images, containing 118K training images and 5K validation images. (2) *ImageNet-100* is a 100-class subset of ImageNet [30] comprising ~ 130 K images across diverse object categories. For controlled experiments, we use a 50-class subset by default and vary the number of classes (from 1 to 100) in ablation studies. (3) *WikiArt* [25] contains 81,444 paintings spanning 27 styles and 45 genres

Table 4: Main results of ImageAuditor on IRAG for Q&A tasks. Following the literature [6], the system uses CLIP ViT-L/14-336 for retrieval and Qwen2.5-VL-7B-Instruct for generation. All methods share the same extraction segment (“Describe each retrieved image...”) and differ only in their retrieval segment, so the results reflect optimizer quality. PoisonedRAG-B* outperforms the two gradient-based variants as it avoids noisy gradient estimation in this challenging setting.

Attack Method	MMQA			MSCOCO		
	AUC	ACC	TPR _{5%}	AUC	ACC	TPR _{5%}
Naïve	N/A			N/A		
PoisonedRAG-B*	0.62	0.62	0.19	0.57	0.59	0.14
PoisonedRAG*	0.53	0.57	0.06	0.51	0.56	0.06
BadRAG*	0.52	0.55	0.04	0.49	0.54	0.03
ImageAuditor	0.85	0.82	0.56	0.82	0.76	0.39

Table 5: Stability of our evaluation. MSCOCO and MMQA are used for T2I and Q&A tasks.

Task	# Tests	AUC	ACC	TPR _{5%}
T2I task	Default	0.91	0.85	0.55
	800	0.91	0.85	0.56
Q&A task	Default	0.85	0.82	0.56
	800	0.85	0.80	0.49

Table 6: Attack evaluation on T2I tasks under both per-dataset and combined settings.

Setting	Dataset	AUC	ACC	TPR _{5%}
Separate	MSCOCO	0.91	0.85	0.55
	WikiArt	0.87	0.82	0.49
	Dogs	0.90	0.84	0.65
Combined	Full	0.87	0.79	0.46

from over 2,000 artists across centuries of art history. We exclude paintings labeled as unknown artists, yielding a filtered dataset of $\sim 39K$ images. We then split the dataset by artist at a 4:1 ratio. (4) *Stanford Cars* [15] consists of 16,185 images of 196 car models, with an official 8,144/8,041 train/test split. (5) *Stanford Dogs* [7] contains 20,580 images of 120 dog breeds, with an official 12,000/8,580 train/test split. (6) *CelebA-HQ* [13] is a high-quality facial dataset containing 30,000 face images at 1024×1024 resolution. We split the dataset at a 4:1 ratio. In total, we construct an image database of 259,057 images spanning all six datasets. Together, they provide concepts covering *real-world scenes*, *fine-grained and rare objects*, *artistic styles*, and *high-quality faces*, etc.

Q&A dataset. (1) *MMQA* (MultiModalQA) [34] is a multi-modal question answering benchmark requiring reasoning across text, tables, and images. We use the complete image partition of the dataset, containing 58,075 images. Due to the lack of official splits, we randomly sample 1,000 images as the non-member set. (2) *MSCOCO* is also used in RagVL [6] as a Q&A dataset to evaluate text-to-image retrieval performance over a large-scale image database.

Utility evaluation. The T2I database covers diverse visual concepts spanning real scenes, fine-grained and rare objects, and artistic styles. For datasets with clear class labels, we randomly sample 20 classes and generate 10 prompts per class—for example, *A photo of a {class} sitting on the ground*” for *Stanford Dogs*, and *A modern artwork influenced by {class}*” for *WikiArt*. This allows us to evaluate whether generated images faithfully capture the visual concepts from each class and to compare generators with and without IRAG. For *MSCOCO* and *CelebA-HQ*, we randomly sample 200 prompts following their respective distributions and compare the generated image distributions using FID and CLIP scores. For *MMQA*, which supplies visual knowledge for knowledge-dependent question answering, we follow the same setup as *AQUA* and measure response accuracy to compare generators with and without IRAG.

D.2 Description of the Generation Settings

For T2I tasks, we use four generators that span different architectures and conditioning mechanisms: (1) *SDXL + IP-Adapter* [26, 40]: We pair SDXL with the ViT-H IP-Adapter Plus variant (ip-adapter-plus_sdxl_vit-h) and use an IP-Adapter scale of 0.6, which provides a balanced setting between image and text conditioning. Images are generated at 1024×1024 resolution with 30 denoising steps, following the default setting of SDXL. Empirically, ImageAuditor maintains AUROC above 80% across practical scale values (≥ 0.5); lower scales weaken retrieval influence and are rarely used in IRAG systems, as they reduce the benefit of retrieval augmentation. (2) *SD1.5 +*

Table 7: IRAG improves the utility of text-to-image generation. Following [22], we use CLIP-I as a metric of concept-level semantic alignment, which measures the similarity between the generated image and the corresponding class prototype embedding.

Generator	RAG	WikiArt			ImageNet100			Stanford Dogs		
		CLIP-I	CLIP-T	FID	CLIP-I	CLIP-T	FID	CLIP-I	CLIP-T	FID
SDXL+IP-Adapter	×	0.453	0.314	330	0.649	0.343	145	0.430	0.324	235
	✓	0.522	0.296	292	0.744	0.358	103	0.623	0.269	138
SD1.5+IP-Adapter	×	0.419	0.292	340	0.617	0.312	189	0.412	0.286	243
	✓	0.539	0.295	297	0.732	0.321	108	0.621	0.264	135

Generator	RAG	Cars			MSCOCO			CelebA-HQ		
		CLIP-I	CLIP-T	FID	CLIP-I	CLIP-T	FID	CLIP-I	CLIP-T	FID
SDXL+IP-Adapter	×	0.618	0.333	160	/	0.362	190	/	0.365	171
	✓	0.688	0.334	112	/	0.364	178	/	0.375	117
SD1.5+IP-Adapter	×	0.608	0.311	124	/	0.343	180	/	0.351	112
	✓	0.684	0.319	111	/	0.347	175	/	0.364	96

Table 8: IRAG improves the utility of question answering. We report accuracy (ACC) on MMQA across different numbers of retrieved passages.

Generator	None	$K = 2$	$K = 3$
Qwen2.5-7B-Instruct	0.314	0.529	0.495
LLaVA-v1.6-Mistral-7B	0.359	0.575	0.573

IP-Adapter [29, 40]: We pair Stable Diffusion 1.5 with its corresponding IP-Adapter Plus variant and use an IP-Adapter scale of 0.6. Images are generated at 512×512 resolution with 30 denoising steps, following the default setting of SD1.5. (3) *Kandinsky v2.2* [28]: We use Kandinsky v2.2, a latent diffusion model with a CLIP-based image prior that natively supports image-conditioned generation. Images are generated at 512×512 resolution with 100 denoising steps, using an image weight of 0.7, a guidance scale of 4.0, a prior guidance scale of 4.0, and 25 prior inference steps. (4) *SDXL + Conceptrol* [11]: We use SDXL augmented with Conceptrol, an advanced image conditioning method. Images are generated at 1024×1024 resolution with 30 denoising steps, using a condition scale of 1.0, a guidance scale of 6.0, and a warmup ratio of 0.2.

For Q&A tasks, we use two vision-language models (VLMs): (1) *Qwen2.5-VL* [37]: We experiment with Qwen2.5-VL, a vision-language model from the Qwen family with native dynamic resolution support. (2) *LLaVA-1.6-Mistral-7B* [19]: We also use LLaVA-1.6-Mistral-7B, a 7B-parameter vision-language model built on Mistral that supports dynamic high-resolution image inputs. All other hyperparameters follow the default settings from each model’s official implementation.

D.3 Description of the Evaluation Metrics

For attack evaluation, we use standard MIA metrics, including AUROC, ACC, and TPR@5%FPR. AUROC measures the area under the ROC curve and captures the overall separability between members and non-members across all thresholds. ACC reports the classification accuracy at the optimal decision threshold. TPR@5%FPR measures the true positive rate when the false positive rate is constrained to 5%, reflecting performance under a strict false-alarm budget. We additionally report the Retrieval Success Rate (RSR) to measure the attack’s effectiveness at retrieving member images. For utility evaluation, we report CLIP-T, CLIP-I (key metric for class-labeled dataset), and FID for T2I tasks and response accuracy for Q&A tasks, following the literature [22, 6].

D.4 Description of the Attack Baselines

Traditional text-RAG MIAs [23, 20] are not directly applicable to our setting due to the incompatibility of T2I generators with question-answering tasks and the inability to embed images into text queries. We therefore adapt these baselines following the query decomposition paradigm of *ImageAuditor*. The naïve baseline includes only the extraction segment: we generate candidate captions and select

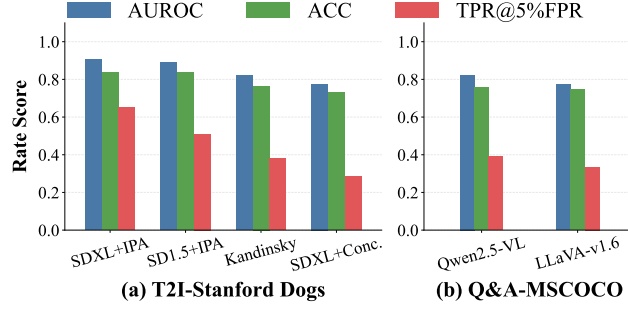


Figure 9: ImageAuditor remains effective across different generators for both T2I and Q&A tasks. SDXL+Conceptrol reduces TPR@5%FPR due to its defensive and more creative generation behavior. However, it still achieves around 30% of TPR@5%FPR, which significantly outperforms random guessing (5%) and is considered an effective attack under standard MIA literature.

Table 9: Impact of extraction segment type on ImageAuditor against IRAG for the T2I task. BLIP and Qwen-VL captions yield stronger attacks, as better alignment between the image and the extraction segment more effectively elicits copy-like behavior from T2I generators.

Extraction Segment Type	MSCOCO			WikiArt			Stanford Dogs		
	AUC	ACC	TPR _{5%}	AUC	ACC	TPR _{5%}	AUC	ACC	TPR _{5%}
Generic Cap.	0.71	0.70	0.38	0.75	0.72	0.39	0.83	0.76	0.49
BLIP Cap.	0.86	0.82	0.42	0.87	0.80	0.54	0.88	0.83	0.51
Qwen-VL Cap.	0.91	0.85	0.55	0.87	0.82	0.49	0.90	0.84	0.65

those with high reward (e.g., similarity to the test image), following our framework. Since these captions semantically describe the test image, this baseline can still retrieve some member images in the T2I setting, providing a meaningful comparison point.

We additionally adapt PoisonedRAG-B [43], PoisonedRAG [43], and BadRAG [39] as baseline attacks. All three are originally designed for text-RAG poisoning, and we adapt them by treating retrieval segment construction as analogous to their adversarial passage optimization. Specifically, we apply their respective objectives to construct retrieval segments for each test image, while keeping the extraction segment and scoring rule identical to ImageAuditor for a fair comparison. The three methods differ primarily in their optimization objective.

(1) *PoisonedRAG-B* [43] is the gradient-free variant of PoisonedRAG. Rather than optimizing T_{ret} via token-level gradients, it directly constructs the retrieval segment by extracting content from the target sample. In our case, we use the generated image caption as T_{ret} . In the T2I setting, this differs from the Naïve baseline in that Naïve uses no retrieval segment, whereas PoisonedRAG-B* employs two separate captions—one as the extraction segment and another as the retrieval segment.

(2) *PoisonedRAG* [43] employs HotFlip [9] to optimize the retrieval segment T_{ret} such that the embedding of the full query is close to that of the target image. Specifically, given a target embedding $\mathbf{e}_t$, a query encoder $E(\cdot)$, and a fixed extraction segment T_{ext} , the optimization objective is:

$$T_{\text{ret}}^* = \arg \max_{T_{\text{ret}}} \text{sim}(E(T_{\text{ext}} \oplus T_{\text{ret}}), \mathbf{e}_t), \quad (2)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and $\oplus$ denotes concatenation. At each iteration, HotFlip selects a random token position and replaces the token with the candidate yielding the largest gradient-based score improvement.

(3) *BadRAG* [39] adopts a contrastive optimization objective (COP) that distinguishes the target from a set of shadow (negative) samples. Let $\mathbf{q} = E(T_{\text{ext}} \oplus T_{\text{ret}})$ denote the query embedding and $\{\mathbf{e}_n^{(i)}\}_{i=1}^N$ denote shadow embeddings drawn from the shadow set. The InfoNCE-style objective is:

$$T_{\text{ret}}^* = \arg \max_{T_{\text{ret}}} \log \frac{\exp(\text{sim}(\mathbf{q}, \mathbf{e}_t)/\tau)}{\exp(\text{sim}(\mathbf{q}, \mathbf{e}_t)/\tau) + \sum_{i=1}^N \exp(\text{sim}(\mathbf{q}, \mathbf{e}_n^{(i)})/\tau)}, \quad (3)$$

Table 10: The shadow dataset improves results when the target embedding space is unknown. Here, ViT-L/14 is used as the shadow embedding model.

Type	MSCOCO			WikiArt			Stanford Dogs		
	AUC	ACC	TPR _{5%}	AUC	ACC	TPR _{5%}	AUC	ACC	TPR _{5%}
No	0.75	0.70	0.31	0.64	0.64	0.18	0.72	0.68	0.26
Yes	0.78	0.75	0.32	0.70	0.67	0.34	0.75	0.70	0.33

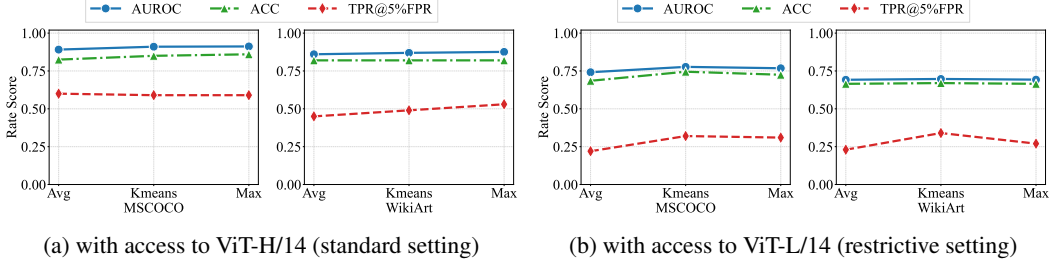


Figure 10: K-means-based aggregation yields the most stable attack results across various settings.

where τ is a temperature hyperparameter. This formulation encourages the retrieval segment to be selectively retrieved for the target while avoiding retrieval for negatives. The optimization is solved using HotFlip-style token-level updates.

D.5 Hyperparameter Sensitivity

The attacker is allowed $M \times N$ queries per test image, where M is the number of extraction segments and N is the number of retrieval segments per extraction segment. We set $M = N = 2$ by default to balance attack effectiveness against query and optimization costs. Using multiple extraction and retrieval segments reduces sensitivity to any single choice. As shown in Figure 11, larger values yield only marginal gains and even a single query (i.e., $M = N = 1$) already achieves competitive performance. For the extraction segment, we use Qwen-VL to generate and rank 5 candidate captions in the T2I setting, and manually design structured prompts in the Q&A setting. For the retrieval segment, we set the token length to 8, which is sufficient for the attack to saturate (see Figure 5(c)). We further sample a small shadow set of 32 images (disjoint from the database) for the contrastive reward, which is practical to collect from public sources. We optimize each retrieval segment for 20 iterations in the T2I setting and 80 iterations in the Q&A setting, due to differences in extraction prompt designs. As shown in Figure 12, our designs provide a favorable efficiency–effectiveness trade-off, with diminishing returns beyond this point. The remaining RGPO parameters take standard values from the policy-optimization literature: $\tau = 1$ (a neutral temperature), $\rho = 0.2$ (a common choice in importance sampling methods), and $\beta = 0.5$ (a balanced adaptation rate). We found these defaults to be robust across all datasets, and therefore did not perform dedicated sweeps over them.

D.6 Description of the Attack Prompts.

We use the following prompts in our attack pipeline:

Prompt for Extraction Segment Generation in the T2I Setting

You are an image captioning system. Given an input image, generate 5 concise and diverse captions.

Requirements:

- Write all captions in English only. Do not use any other language.
- Each caption should describe the main scene and key objects.
- Keep each caption short and natural (similar to MSCOCO style).
- Use different wording and focus on slightly different aspects (e.g., objects, actions, relationships, or context).
- Avoid repeating the same sentence structure or phrases.
- Do not include unnecessary details or long descriptions.

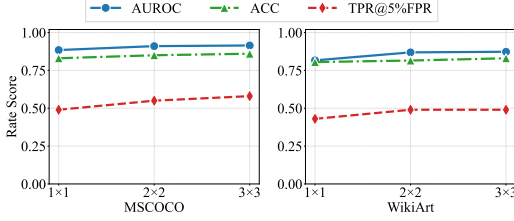


Figure 11: Increasing the number of queries ($M \times N$) per test case slightly improves the attack performance of ImageAuditor. Our attack remains effective with a single query due to the RGPO mechanism.

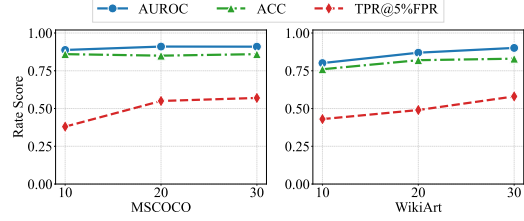


Figure 12: Increasing the number of RGPO iterations slightly improves the results (efficiency vs. effectiveness trade-off), and ImageAuditor remains effective even with a small number of iterations (e.g., 10).

Table 11: In both standard and restrictive settings, a stronger scoring model improves attack performance, as it more accurately captures fine-grained image semantics.

Dataset	Scoring model	with access to ViT-H/14			with access to ViT-L/14		
		AUC	ACC	TPR _{5%}	AUC	ACC	TPR _{5%}
MSCOCO	EvaCLIP	0.87	0.85	0.32	0.78	0.75	0.32
	CLIP ViT-H/14	0.91	0.85	0.55	0.82	0.78	0.39
WikiArt	EvaCLIP	0.81	0.77	0.32	0.68	0.68	0.27
	CLIP ViT-H/14	0.87	0.82	0.49	0.69	0.71	0.41
Dogs	EvaCLIP	0.88	0.82	0.53	0.75	0.70	0.33
	CLIP ViT-H/14	0.90	0.84	0.65	0.77	0.73	0.41

Output format:

1. ...
2. ...
3. ...
4. ...
5. ...

Extraction Segment in the Q&A Setting

Describe each retrieved image. Format as 'Image 1:', 'Image 2:', etc.

Prompt for Details Extraction in the Q&A Setting

You are an image captioning system.

Given an input image, write one detailed natural caption.

Requirements:

- Describe visible objects, text, scene, attributes, and layout when helpful.
- Stay faithful to the image.
- Do not speculate beyond what is visible.
- Do not output numbering, explanations, or extra text.
- Output exactly one caption.

E Discussion

E.1 Generalization to multi-modal inputs.

Our attack also extends to IRAG systems that take image-text pairs as input. Specifically, we can use the image as the retrieval segment and the entire text as the extraction segment, which will further improves attack performance due to the continuous nature of the pixel space. More broadly, this decomposition applies to other multimodal RAG variants, suggesting that the vulnerability is fundamental to cross-modal retrieval.



Figure 13: Visual illustration of ImageAuditor on test images from the MSCOCO dataset under IRAG for the T2I task. Our method performs multiple queries and applies K-means clustering to the resulting MIA scores for selective aggregation.

E.2 Responsible Deployment of ImageAuditor

ImageAuditor should be deployed with care to avoid false accusations. First, because the MIA score distribution varies slightly across datasets (see Table 6), auditors should calibrate the decision threshold on a small set of known non-members from the same domain and select an operating point with a controlled false-positive rate (e.g., $\text{FPR} \leq 5\%$ or even $\text{FPR} \leq 1\%$). Second, single-query decisions can be noisy; auditors should issue multiple queries and apply the K-means aggregation in Sec. 3.4 to retain only high-confidence ones, treating low-agreement cases as inconclusive. Finally, MIA outputs provide statistical evidence rather than definitive proof, and we recommend using ImageAuditor as a screening tool to flag candidates for further investigation rather than as the sole basis for accusation.



Figure 14: Visual illustration of ImageAuditor on test images from the WikiArt dataset under IRAG for the T2I task. Our method performs multiple queries and applies K-means clustering to the resulting MIA scores for selective aggregation.

F Qualitative Analysis

Figure 13–15 visualize our attack results on the MSCOCO, WikiArt, Dogs datasets under the T2I setting, and Figure 16 visualizes our attack results on the MMQA dataset under the Q&A setting. Our method effectively distinguishes members from non-members by eliciting distinct generation behaviors. For example, for the member in the first column of Figure 14 (a painting of a Venetian palace), *ImageAuditor* successfully triggers the copy-like behavior of the diffusion model: across all three queries, the generated images closely replicate the reference image’s composition, color palette, and architectural details, yielding consistently high similarity scores (~ 0.85). In contrast, for the non-member in the last column of Figure 14 (a painting of figures in a garden), the generator produces images that share semantic content with the caption (i.e., a mother and daughter sitting in a garden) but differ substantially from the reference in visual composition, lighting, and stylistic details. As a result, the similarity scores remain consistently lower (~ 0.67 – 0.71). This contrast confirms that our caption-based extraction strategy reliably amplifies the membership signal: members trigger near-verbatim reconstruction through the diffusion model’s copy-like behavior, while non-members yield only semantically aligned but visually divergent outputs.



Figure 15: Visual illustration of ImageAuditor on test images from the Stanford Dogs dataset under IRAG for the T2I task. Our method performs multiple queries and applies K-means clustering to the resulting MIA scores for selective aggregation.

G Computation Resources

All experiments were conducted on three single-GPU systems, each equipped with an NVIDIA A6000 GPU with 48GB of VRAM. We did not use multi-GPU training or distributed execution. Both attack optimization and retrieval-augmented generation were performed on a single GPU per run.

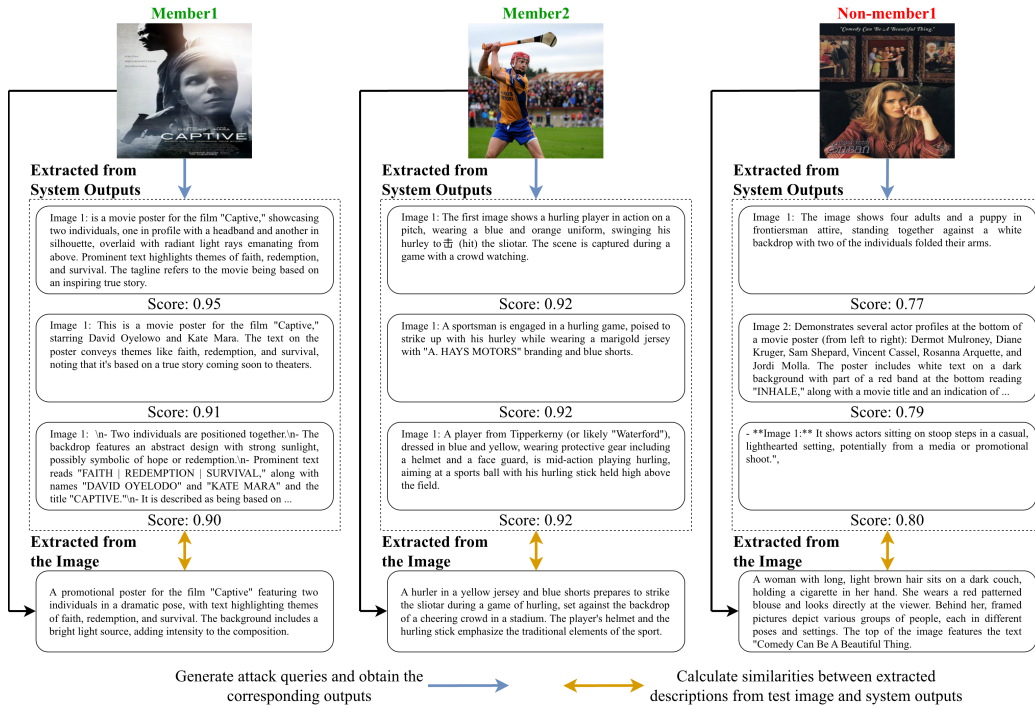


Figure 16: Visual illustration of ImageAuditor on test images from the MMQA dataset under IRAG for the Q&A task. Our method performs multiple queries and applies K-means clustering to the resulting MIA scores for selective aggregation.